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Roadmap for Lessons Learned and Strategies Implemented to Enhance Plug and Perf Fracturing Operations, Leading to Achieving the Highest Stages Per Day

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Abstract

This paper outlines lessons learned and strategies deployed to enhance plug-and-perf completions in Unconventional Oil wells within Abu Dhabi's Shilaif Formation. It elaborates on steps taken to integrate engineering design, fluid optimization, and operational planning to improve stage delivery while minimizing TBS (time-between-stages), a measure critical for UC Wells delivery and monetization. The paper highlights year-on-year progression and advancements in efficiency, executing the highest UC frac stages per day.

Minimizing time-between-stages (TBS) for accelerated unconventional frac well delivery was achieved through zipper fraccing across four wells maximizing efficiency, improved rig-up/rig-down times with zipper frac manifold and high-pressure flex hoses, and a reliable supply chain for continuous sand and water supply. Non-productive time and maintenance were reduced using split stream pumping. Preloaded shorter perforating guns and all-composite / dissolvable frac plugs enabled faster pump-down maximizing Wireline Plug and Perf efficiency. Job design tailored to UAE formations, based on industry lessons learned, further boosted per day stage delivery. This paper provides insights into these innovations and their impact on operational efficiency.

The repair and maintenance time for high-pressure pumps was significantly reduced by 50% through the successful implementation of Split Stream Operations combined with a High Vis Friction Reducer. The use of a Zipper Frac Manifold and high-pressure flex hoses reduced the Rig Up/Down time by 30% through reducing number of components and connections avoiding the maze of multiple iron strings and numerous twists, turns, and connections. The deployment of dissolvable plugs post-fracturing mill-outs was minimized, with most wells being milled out successfully within 48 hours. All-composite plugs demonstrated durability in high-pressure during fraccing. Continuous water delivery was ensured through the construction of two water pits with a combined capacity of up to 200,000 barrels and on-site drilling of shallow water wells. A dedicated sand storage area and loading plant were temporarily built close to the site for continuous sand supply, complemented by sand silos reloading with bulkers. PAD execution was coordinated by creating command center on-site for clear communication and assigning dedicated crews with well-defined tasks in supply chain. Coordination of frac design, fluid selection, perforating technology,

plug strategy, and logistics proved instrumental in achieving high-intensity fracturing of up to 10 stages per day significantly reducing TBS by 50%.

Achieving 10 stages per day was made possible by minimizing time between stages through the deployment of new technologies. These included Zipper Frac operations using a Zipper Frac Manifold, Split Stream Operations to reduce repair and maintenance time, and high-pressure flex hoses to minimize rig up/down time. Additionally, the use of dissolvable frac plugs allowed for faster mill outs. Robust logistics and supply chain management ensured a continuous supply of sand and water.

Introduction

The development of unconventional oil resources in the United Arab Emirates (UAE) represents a change in the region's upstream strategy. While the industry has historically focused on conventional reservoirs, operators in Abu Dhabi are now exploring tighter formations such as the Shilaif Formation to reach production goals and diversify their resources. This shift has led to the use of advanced completion techniques, with plug-and-perf (PnP) fracturing being widely implemented due to its flexibility and cost considerations.

A key performance metric in unconventional completions is Time Between Stages (TBS)—the interval between the end of one fracturing stage and the start of the next. Minimizing TBS is critical not only for reducing non-productive time (NPT) but also for accelerating well delivery and early monetization. As one study noted, "reducing TBS is not just a matter of operational speed; it's a strategic imperative for maximizing asset value in unconventional plays."1 This principle guided the operational philosophy in the Shilaif campaign, where achieving up to 10 stages per day became a benchmark of efficiency.

The Shilaif Formation presents unique challenges, including variable stress regimes, tight lithologies, and logistical constraints. To overcome these, the project team implemented a suite of innovations across engineering design, execution strategy, and supply chain management. Engineering enhancements included tailored job designs based on local geomechanics, the use of preloaded shorter perforating guns to reduce wireline time, and the deployment of dissolvable and all-composite frac plugs to streamline mill-out operations.2

Operational strategies were equally transformative. Zipper fracturing across multiple wells enabled simultaneous operations, significantly reducing idle time. Split stream pumping allowed for concurrent equipment maintenance and fracturing, cutting repair time by half3. The use of high-pressure flex hoses and a zipper frac manifold simplified rig-up/down processes, reducing transition time by 30% and eliminating the complexity of traditional iron strings.4

Logistics played a pivotal role in sustaining high-intensity operations. Continuous water supply was ensured through the construction of large-capacity water pits and shallow wells, while sand delivery was optimized via on-site storage and loading infrastructure. A centralized command center facilitated real-time coordination among crews, ensuring seamless execution and rapid troubleshooting.5

These integrated strategies culminated in a record-setting performance, with consistent delivery of up to 10 stages per day. This paper presents a roadmap of the lessons learned and technologies deployed, offering insights for replication in similar unconventional environments across the Middle East and beyond.

Shilaif Formation

The Shilaif Formation, situated onshore in Abu Dhabi, represents a Cenomanian-aged carbonate source rock deposited within a deeper water intrashelf basin environment. It is contemporaneous with the neighboring Mishrif Formation, which formed under shallower, higher-energy conditions. The Shilaif Formation comprises three principal stacked sequences, each containing both organic-rich source rock and dense, low-porosity limestone, thereby functioning as a dual-purpose reservoir with both hydrocarbon generation and production potential.6

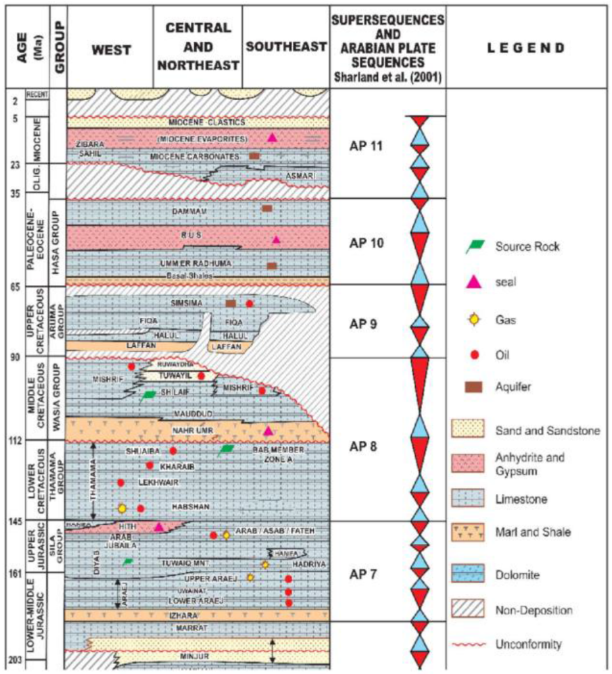


Figure 1—Simplified stratigraphic sequence and hydrocarbon habitat distribution of Abu Dhabi petroleum systems (Alsharhan et al., 2014)

In the western UAE, the Shilaif Formation is recognized as one of the main source rocks of the Cretaceous. Its lithology includes argillaceous limestone, lime mudstone, and shale. Geochemical analysis and basin modelling have shown high TOC, large thickness, and high maturity, particularly in the southeastern part of the basin. These characteristics make it a high-potential area for unconventional oil exploration.7

Advanced seismic and petrophysical techniques have been employed to predict sweet spots in the Lower Shilaif, including 3D seismic CRP gather conditioning, pre-stack inversion, fracture prediction, and in-situ stress analysis. These workflows have enabled the identification of zones with high brittleness, high pressure, and favourable fracture characteristics, which are essential for successful stimulation and production.7

In summary, the Shilaif Formation presents both geological complexity and significant resource potential. Its successful development hinges on robust subsurface integration, adaptive completion technologies, and continuous refinement of operational strategies.

The reservoir characteristics indicate that intervals are well-suited for stimulation through hydraulic fracturing, provided that fluid systems and execution plans are specifically tailored to unlock their production capabilities. Fig. 2.

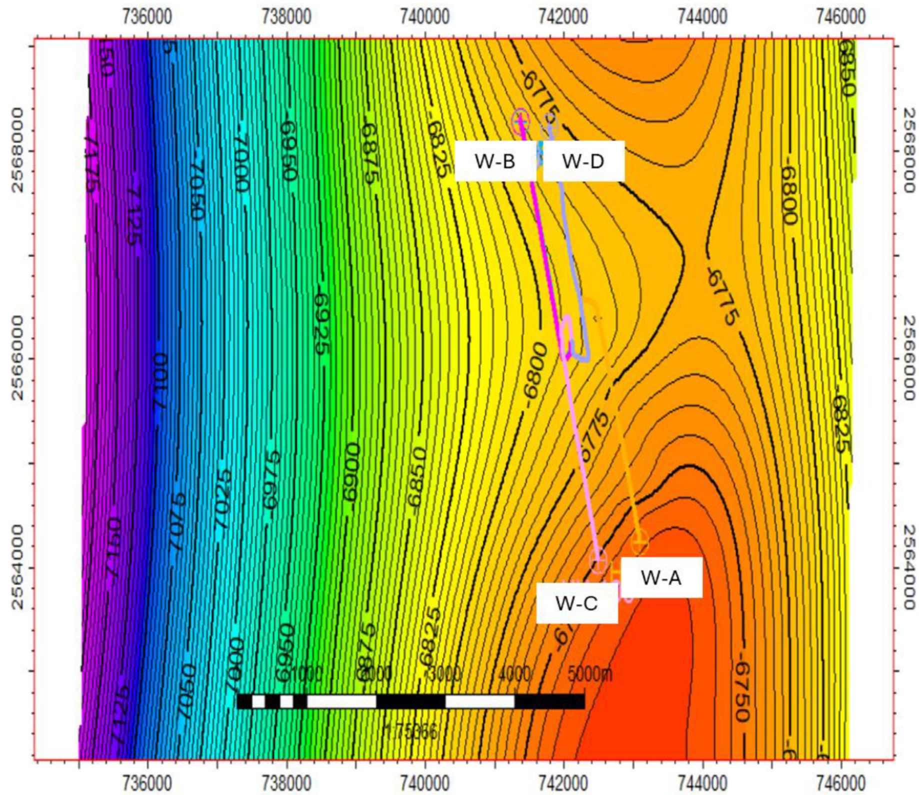


Figure 2—Structure Map, Wells A, B, C & D.

PAD Strategy

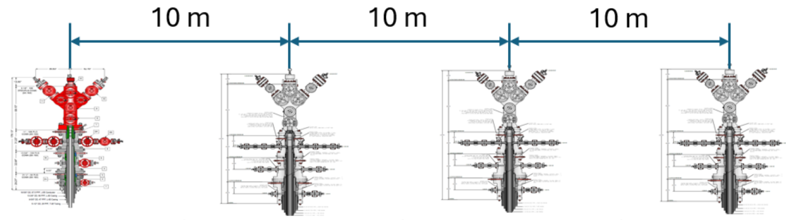
A comprehensive execution framework was deployed to streamline well stimulation and completion activities by integrating hydraulic fracturing, wireline interventions, and coiled tubing operations into a unified workflow. This multi-service coordination was designed to boost operational throughput, minimize downtime, and enhance reservoir productivity through synchronized task execution and adaptive decision-making.

A key innovation in this framework was the implementation of split-stream fracturing, which enabled concurrent stimulation of multiple reservoir zones. This technique shortened stage durations and expanded reservoir exposure. Fracturing treatments were customized with specific equipment setups and validated in the field to optimize proppant delivery and placement. Engineers utilized live data feeds—tracking pressure, flow rates, and fluid dynamics—to fine-tune operations in real time, ensuring effective stimulation across horizontal well sections.

Wireline operations played a pivotal role in the plug-and-perf process, acting as the bridge between subsurface targeting and stimulation success. The workflow was refined to accelerate pump-down procedures, reducing turnaround times and improving stage execution rates. Perforation plans were precisely aligned with reservoir intervals, supported by advanced logging tools that delivered high-resolution subsurface data. These insights allowed for continuous refinement of stage designs and improved fracture initiation reliability.

Following stimulation, coiled tubing was deployed using a sequenced milling approach, featuring high-efficiency bottom hole assemblies (BHAs) tailored to the well's thermal and depth conditions. This setup ensured rapid and complete removal of composite and dissolvable plugs, preventing debris accumulation and maintaining wellbore integrity.

Throughout the campaign, integrated diagnostics and inter-service communication enabled granular performance tracking and responsive adjustments. This collaborative model allowed all service lines to operate in harmony, resulting in a streamlined completion process that reduced complexity and delivered superior stimulation outcomes.



	WELL A	WELL D	WELL C	WELL B
Well type	Horizontal (Oil)	Horizontal (Oil)	Horizontal (Oil)	Horizontal (Oil)
Deviation at EOT	92.33º Inc.	91.69º Inc.	90.72º Inc.	89.60º Inc
Frac String	5 1/2", 26 ppf, T-95	5 1/2", 23 ppf, Q-125	5 1/2", 23 ppf, P-110	5 1/2", 26 ppf, T-95
Well TD	15,208.8 ft.MD (7,882.6 ft.TVD)	14,950.0 ft MD (7,928.9 ft.TVD)	15,100 ft.MD (7,786 ft.TVD)	14,994.96 ft.MD (7907.1 ft.TVD)

Figure 2a—PAD Wells Conductor Layout Sketch and Configuration.

Wells Preparation

Prior to initiating the main stimulation campaign, each well underwent a series of preparatory tasks conducted offline and tailored to its specific location. The readiness sequence followed a deliberate order—starting with Well A, followed by Well D, then Well C, and concluding with Well B. This structured progression enabled efficient deployment of resources and minimized delays across the pad.

Site was equipped with the necessary equipment, underwent safety verifications, and was aligned with service line requirements to support the integrated execution strategy. The offline nature of these activities allowed multiple wells to be prepared in parallel, streamlining transitions between operations and enhancing overall execution flow.

Preparations included coiled tubing cleanouts and wellbore displacement using treated water, cement evaluations via wireline, tree swaps from production to fracturing configurations, and integrity tests at pressures of 13,500 psi for the wellhead and 3,000 psi for the annulus. Perforation activities were initiated with tractor-conveyed wireline runs, and diagnostic fracture injection tests (DFITs) were conducted in Well D & Well B.

This phased and well-orchestrated approach ensured that each well was primed for stimulation with minimal disruption, laying the groundwork for a high-efficiency multi-well fracturing campaign.

PAD Execution Summary

A hydraulic fracturing campaign comprising four wells—A, D, B, and C—was implemented utilizing a zipper frac approach to optimise both equipment usage and crew productivity. Each well was directed at the Shilaif carbonate reservoir and underwent 30 stages of slickwater fracturing, with 100-mesh sand employed as the primary proppant. Acid spearheads (15% HCl) were selectively utilized, typically introduced during Stage 1 and, in certain cases, in subsequent stages.

Wells A and D were stimulated first, both completed with 5½" frac strings. They operated at ~100 bpm and ~7,500 psi. Well A used dissolvable plugs early and composite plugs later, while Well D incorporated three plug types and faced a wireline cut incident requiring fishing. Despite this, both wells completed successfully and transitioned to coiled tubing milling.

Wells B and C followed, with similar designs and acid treatments in Stages 1 and 30. Fracturing pressures averaged ~7,353 psi (B) and ~7,496 psi (C). Plug configurations included composite and all-composite plugs. Minor issues like misfired guns and weather delays were resolved without major impact, and both wells proceeded to testing after milling.

This method provided consistent results across the pad and resolved well-specific issues, highlighting zipper fracturing's effectiveness for multi-well developments in the Shilaif Formation. Table 1.

Table 1—PAD Wells Planned Activities Sequence.

Activity Sequence	Well A	Well D	Well C	Well B
1	CT Wellbore Cleanout			
2	CBL	CT Wellbore Cleanout		
3	Tree Swap X-mas Tree →Frac Tree	CBL	CT Wellbore Cleanout	
4	Integrity Test	Tree Swap X-mas Tree →Frac Tree	CBL	CT Wellbore Cleanout
5		Integrity Test		CBL
6		WL Perforate Stg-1 Tractor RIH		
7	WL Perforate Stg-1 Tractor RIH	DFIT		
8	Zipper Frac Frac / PnP Stage 1 - 30	Zipper Frac Frac / PnP Stage 1 - 30		
9	Tree Swap Frac Tree → X-mas Tree	Tree Swap Frac Tree → X-mas Tree		Tree Swap X-mas Tree →Frac Tree
10			Tree Swap X-mas Tree →Frac Tree	Integrity Test
11			Integrity Test	WL Perforate Stg-1 Tractor RIH
12			WL Perforate Stg-1 Tractor RIH	DFIT
13			Zipper Frac Frac / PnP Stage 1 - 30	Zipper Frac Frac / PnP Stage 1 - 30
14				CT Plugs Millout
15			CT Plugs Millout	
16		CT Plugs Millout		
17	CT Plugs Millout			

Methodology and Execution Roadmap

Delivering high-efficiency Multi-Stage Frac completions in the Shilaif Formation required an integrated, end-to-end execution plan addressing the technical, operational, and logistical complexities of unconventional development in Abu Dhabi. The methodology was built around three interdependent pillars: Engineering Design Enhancements, Operational Strategy Execution, and Logistics and Supply Chain Optimization. Each pillar was systematically developed to achieve a key objective—maximize stage count per day while minimizing time-between-stages (TBS) and non-productive time (NPT).

Engineering Design Enhancements

The first step was to align stimulation design with the Shilaif Formation's unique rock mechanics and completion constraints. This was accomplished via:

Tailored Fracture Job Design.

- Integrated geomechanical models, stress profiling, and brittleness analysis to optimize stage length, cluster spacing, and fluid volumes.
- Designs were iteratively updated based on real-time microseismic and pressure response feedback, enhancing cluster efficiency and reducing frac hits.
- Improved Fracture Conductivity: Higher concentrations improved fracture width and ensured long-term production sustainability. Increased per-stage effectiveness reduced the need for excessive stage counts in some laterals leading to time and cost savings by reduced proppant delivery time per well by optimizing sand volume against reservoir needs.

Preloaded, Short-Length Perforating Guns.

- Shorter guns preloaded with optimized charges allowed rapid deployment and minimized wireline rig-up/down cycles.
- Reduced reload frequency accelerated stage turnover and improved safety by limiting tool handling.

All-Composite and Dissolvable Frac Plugs.

- Composite plugs provided the high-pressure sealing required for zonal isolation during high-rate fracs.
- Dissolvable plugs enabled efficient post-job cleanouts, with the majority of wells milled out within 48 hours post-stimulation, cutting post-frac wait times by over 40%.

Operational Strategy Execution

To maintain continuous pumping and maximize asset productivity, an integrated multi-well execution strategy was deployed.

Zipper Frac Across two Wells at a time.

- Executed a 2-well zipper frac sequence using dedicated manifolds, enabling simultaneous prep and pumping across wells.
- This sequencing reduced idle time and increased daily stage count by ~50% compared to single-well operations. Split Stream Pumping
- Deployed a dual-stream pumping system where one stream maintained dirty fluid flow and the other clean fluid, reducing the stress in the lines and pumping units and maintenance downtime. Allowed proactive maintenance rather than reactive repairs, minimizing critical failures.
- This configuration reduced high-pressure pump repair downtime by 50%, and friction reducer chemistry was adjusted to ensure consistent proppant transport in both streams.

High-Pressure Flex Hose and Zipper Frac Manifold.

- Replaced traditional iron strings with flex hoses, eliminating time-intensive rig-ups and minimizing iron-related risks.
- The manifold setup reduced connection complexity, lowering the potential for leaks and mechanical failures while reducing rig-up/down time by approximately 30%. Fig. 3.

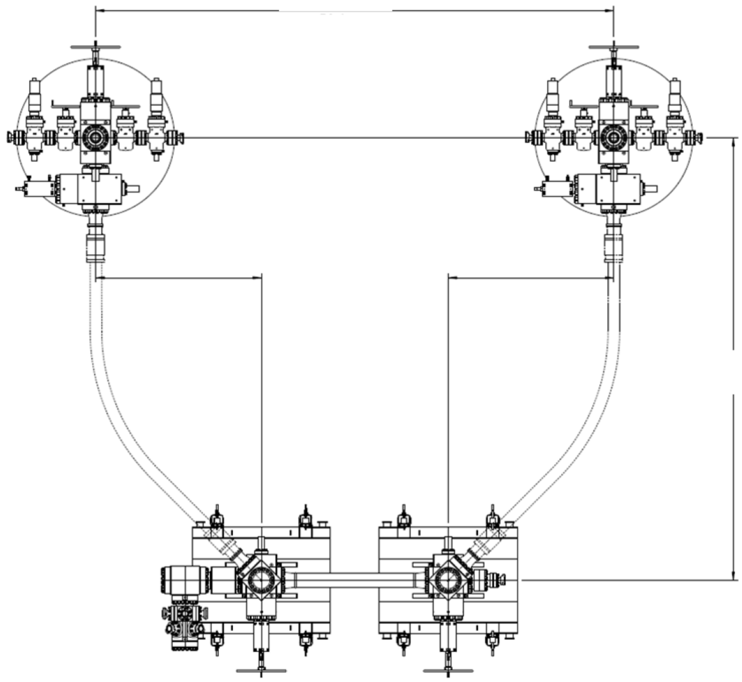


Figure 3—Zipper Manifold General Diagram.

Logistics and Supply Chain Optimization

Uninterrupted resource availability and a streamlined coordination framework were pivotal to sustaining high-intensity operations.

Water Supply Assurance.

- Two engineered water pits (~200,000 bbl total capacity) were constructed near the well pad.
- Shallow water wells were drilled to supplement supply, ensuring consistent hydraulic horsepower delivery without reliance on external tankers.

This resulted in a steady water supply, eliminating frac pauses from shortages—especially during high-rate zipper fracs on multiple wells. It also provided supply redundancy during extreme weather or disruptions, while cutting water trucking needs to improve pad safety and reduce congestion.

Proppant Storage and Handling.

- A dedicated sand logistics hub was temporarily constructed adjacent to the pad, featuring:
  - Bulk storage silos continuously replenished by trucked-in bulkers.
  - A close by sand loading facility to reduce transportation lag and mitigate weather or traffic-related delays.
  - Minimized dependency on just-in-time deliveries over long distances.

On-Site Command and Control Center

A central field operations command center was deployed to manage all fracturing activities in real-time. Benchmarking and KPIs: Enabled precise tracking of stage performance, facilitating post-job analysis and continuous improvement.

- Coordinated job sequencing, crew rotation, and equipment dispatch were managed through a digital dashboard, allowing proactive response to changing conditions and minimizing operational inefficiencies.

This approach lead to improved decision making, enabled real-time adjustments to stage execution, helping to maintain daily targets. Additionally this allowed for rapid troubleshooting for which live visibility into anomalies such as pressure spikes or equipment lag helped avoid job stoppages.

Execution Roadmap

To translate this methodology into field operations, a phased roadmap was developed:

Table 2—Execution Roadmap Table.

Phase	Activity	Key Outcome
Phase 1	Detailed subsurface study and well selection	Optimized candidate wells for zipper fracs
Phase 2	Equipment and infrastructure mobilization	Reduced setup times via modular designs
Phase 3	Engineering design customization	Improved fracture efficiency via tailored plans
Phase 4	Zipper frac & split stream execution	Achieved record 10 stages/day with 50% lower TBS
Phase 5	Post-job evaluation and plug mill-out	<48 hr cleanout for most wells
Phase 6	Continuous improvement loop	Lessons learned captured and applied to next PAD

The ongoing improvement of the observed elements and their implementation contributed to measurable progress during pad execution, ultimately resulting in significant downtime reduction and enhanced overall well completion efficiency.

Operational Results and Field Activities Enhancement

A multi-well hydraulic stimulation campaign was carried out on four horizontal wells—A, D, B, and C—using a zipper fracturing technique that allowed concurrent operations and optimized crew and equipment utilization. Each well targeted the Shilaif tight carbonate reservoir and underwent a 30-stage slickwater fracturing sequence. Despite encountering operational hurdles such as wireline tool issues and weather-related interruptions, all wells were successfully completed and transitioned into coiled tubing milling and testing phases. The consistent execution across the pad and strategic design choices led to measurable improvements in efficiency, setting new benchmarks in completion performance and stimulation quality.

These results were driven by several innovations. Split-stream fracturing enabled simultaneous fluid and proppant delivery through multiple channels, increasing pump rates and reducing friction losses. This approach minimized downtime for equipment maintenance and accelerated stage turnover. The use of high-viscosity friction reducers (HVFRs) allowed for higher proppant loading while maintaining low friction, with fluid performance validated through rigorous rheological testing and tailored dosing strategies.

Efficiency gains were driven by innovations such as split-stream fracturing, high-viscosity friction reducers, and streamlined sand logistics, which enabled up to nine stages per day and reduced completion time per well by 56%. Wireline operations saw a 33% boost in efficiency through the adoption of plug-and-play gun systems and vertical handling methods, while coiled tubing milling achieved a 59% reduction in valve-to-valve duration through optimized BHAs and real-time diagnostics. These integrated enhancements not only improved individual well outcomes but also enabled a scalable, cost-effective model for multi-well pad development, accelerating time-to-market and maximizing reservoir contact.

The integrated execution model proved instrumental in unlocking the production potential of the Shilaif unconventional reservoir. By coordinating hydraulic fracturing, plug-and-perf, and post-frac cleanout operations under a unified framework, the team achieved significant reductions in cycle time and non-productive time. This approach enabled a shift from exploratory single-well operations to scalable multi-well pad development, accelerating time-to-market and enhancing reservoir contact.

Conclusion

The outlined methodology and roadmap collectively enabled a leap in execution efficiency for unconventional completions in the Shilaif Formation. By integrating innovative engineering design, agile operational strategies, and robust logistics planning, a new benchmark of 10 stages per day was achieved. These efforts significantly accelerated the time to production while reducing costs, and the lessons learned offer a replicable model for other unconventional plays globally.

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Common Abbreviation and Units Used

°F	Fahrenheit grades
bbl	barrels
BHA	bottomhole assembly
BHP	bottomhole pressure
BHT	bottomhole temperature
bpm	barrels per minute
CBL	cement bond log
CSG	basing
CT	coiled tubing
DFIT	diagnostic fracture injection test
DHWS	down hole well status
FC	float collar
FS	frac string
ft	feet
GORV	gas operated relief valve
gpt	gallon per thousand
GR	gamma ray
H_2S	hydrogen sulfide
HCl	hydrochloric acid
HP	high pressure
HVFR	high viscous friction reducer
ID	inner diameter
in	inches
Inc	inclination
LP	landing point
MD	measured depth
min	minutes

OD	outside diameter
PnP	plug and perf
POOH	pull out of hole
ppf	pound per feet
ppg	pound per gallon
psi	pound square inch
RIH	run in hole
TVD	true vertical depth
WHP	wellhead pressure
XO	cross over

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